Diagnostic test — solutions and marking rubric

Mark each question out of 4 using the general rubric in the README. Under each solution, **Marking** says what earns each level for that question. Then apply the routing table.

Section A — Proofs, logic and sets (module 1.1)

A1

Negation: swap each quantifier and negate the innermost statement. The negation of $P \implies Q$ is $P \wedge \neg Q$.

$$\exists \varepsilon > 0 \; \forall \delta > 0 \; \exists x \in \mathbb{R} : \; |x - a| < \delta \;\text{ and }\; |f(x) - f(a)| \geq \varepsilon.$$

The original says f is **continuous at** a. The negation says f is **discontinuous at** a: there is a fixed tolerance ε such that, however close you look to a, some point x has $f(x)$ at least ε away from $f(a)$.

Marking. 4: correct negation with $\geq$ and "and", plus both interpretations. 3: one slip (for example $>$ instead of $\geq$). 2: quantifiers swapped correctly but the implication negated wrongly. 1: knows it is about continuity only.

A2

Let $P(n)$ be the statement $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

Base case. $n = 1$: left side 1, right side $(1 \cdot 2/2)^2 = 1$.

Inductive step. Assume $P(n)$. Then

$$\sum_{k=1}^{n+1} k^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3 = \frac{(n+1)^2\,(n^2+4n+4)}{4} = \frac{(n+1)^2(n+2)^2}{4},$$

which is $P(n+1)$. By induction $P(n)$ holds for all $n \in \mathbb{N}$.

Marking. 4: base case, explicit use of the hypothesis, correct algebra. 3: algebra slip. 2: structure right but hypothesis not visibly used. 1: knows the template only.

A3

(a) Suppose $f(a) = f(a')$. Applying g gives $g(f(a)) = g(f(a'))$, that is $(g \circ f)(a) = (g \circ f)(a')$. Since $g \circ f$ is injective, $a = a'$. So f is injective.

(b) Let $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, $f(1) = 1$, and $g(1) = g(2) = 1$. Then $g \circ f$ has a one-point domain, so it is injective, but g is not injective.

(c) If $y \in f(A_1 \cap A_2)$ then $y = f(a)$ for some a in both A_1 and A_2, so $y \in f(A_1)$ and $y \in f(A_2)$. Strict example: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, $A_1 = \{-1\}$, $A_2 = \{1\}$. Then $f(A_1 \cap A_2) = f(\emptyset) = \emptyset$, but $f(A_1) \cap f(A_2) = \{1\}$.

Marking. Roughly 4/3 of a point per part; round to the nearest whole number. Any valid counterexample earns full credit.

A4

Reflexive: $u = 1 \cdot u$. *Symmetric:* if $v = \lambda u$ with $\lambda > 0$ then $u = \lambda^{-1} v$ and $\lambda^{-1} > 0$. *Transitive:* if $v = \lambda u$ and $w = \mu v$ then $w = (\mu\lambda) u$ with $\mu\lambda > 0$.

The class of u is $\{\lambda u : \lambda > 0\}$, the **open ray** from the origin through u (the origin itself excluded). Each ray meets the unit circle S^1 in exactly one point, $u/\|u\|$. The map $[u] \mapsto u/\|u\|$ is well defined, because $\lambda u/\|\lambda u\| = u/\|u\|$ for $\lambda > 0$. It is injective, because $u/\|u\| = v/\|v\|$ gives $v = (\|v\|/\|u\|)\, u$. It is surjective, because a point $p \in S^1$ is the image of $[p]$.

Marking. 4: all three properties, rays, a bijection with well-definedness checked. 3: well-definedness not checked. 2: equivalence relation only. 1: partial.

A5

(a) For $n \in \mathbb{N}$ let F_n be the set of subsets of $\{1, \dots, n\}$. It is finite, with 2^n elements. Every finite subset of $\mathbb{N}$ lies in some F_n (take n to be its largest element, or $n = 1$ for $\emptyset$). So the set of finite subsets is $\bigcup_n F_n$, a countable union of finite sets, which is countable.

Alternative: $S \mapsto \sum_{k \in S} 2^{k-1}$ is a bijection from the finite subsets of $\mathbb{N}$ onto $\{0, 1, 2, \dots\}$, by uniqueness of binary expansions.

(b) Cantor's argument. Let $f : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ be any function and set $D = \{n \in \mathbb{N} : n \notin f(n)\}$. If $D = f(m)$ for some m, then $m \in D \iff m \notin f(m) = D$, a contradiction. So no f is surjective, and $\mathcal{P}(\mathbb{N})$ is uncountable.

Marking. 2 points per part. In (a), give 1 point for "countable union of finite sets" without justification.

A6

Schröder–Bernstein. If there are injections $A \to B$ and $B \to A$, then there is a bijection $A \to B$.

Inclusion $(0,1) \to [0,1]$ is injective. The map $[0,1] \to (0,1)$, $x \mapsto \frac{1}{4} + \frac{x}{2}$, is injective, with image $[\frac{1}{4}, \frac{3}{4}] \subset (0,1)$. By Schröder–Bernstein, $|[0,1]| = |(0,1)|$.

(An explicit bijection also earns full marks. One example: send $0 \mapsto \frac{1}{2}$, $1 \mapsto \frac{1}{3}$, $\frac{1}{n} \mapsto \frac{1}{n+2}$ for $n \geq 2$, and fix every other point.)

Marking. 4: correct statement and two injections checked. 3: one injection not checked. 2: statement only, or injections only. 1: vague.

Section B — Real analysis (module 1.2)

B1

Completeness axiom. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.

($\Rightarrow$) Let $s = \sup S$ and $\varepsilon > 0$. Then $s - \varepsilon < s$, so $s - \varepsilon$ is not an upper bound. Hence some $x \in S$ has $x > s - \varepsilon$.

($\Leftarrow$) Let s be an upper bound with the ε property, and let t be any upper bound. If $t < s$, put $\varepsilon = s - t > 0$. Then some $x \in S$ has $x > s - \varepsilon = t$, contradicting that t is an upper bound. So $s \leq t$ for every upper bound t, and $s = \sup S$.

Marking. 1 point for the axiom, 1.5 per direction.

B2

$$\left|\frac{3n+1}{n+2}-3\right| = \frac{5}{n+2} < \frac{5}{n}.$$

Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > 5/\varepsilon$ (possible by the Archimedean property). For $n \geq N$, $\left|\frac{3n+1}{n+2}-3\right| < 5/n \leq 5/N < \varepsilon$.

Marking. 4: correct estimate and an explicit N. 3: N given but the final inequality not shown. 2: correct algebra only.

B3

Bolzano–Weierstrass. Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Let (a_n) be Cauchy.

Bounded: with $\varepsilon = 1$ there is N such that $|a_n - a_N| < 1$ for $n \geq N$. So $|a_n| \leq \max(|a_1|, \ldots, |a_{N-1}|, |a_N| + 1)$ for all n.

Convergent subsequence: by Bolzano–Weierstrass, $a_{n_k} \to a$ for some a.

Whole sequence converges: given $\varepsilon > 0$, choose N with $|a_n - a_m| < \varepsilon/2$ for $m, n \geq N$, and choose k with $n_k \geq N$ and $|a_{n_k} - a| < \varepsilon/2$. For $n \geq N$, $|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \varepsilon$.

Marking. 1 for the statement; 1 each for boundedness, applying Bolzano–Weierstrass, and the final $\varepsilon/2$ argument.

B4

(a) **Converges absolutely.** Ratio test: $\dfrac{(n+1)^2/2^{n+1}}{n^2/2^n} = \dfrac{(1+1/n)^2}{2} \to \frac{1}{2} < 1$.

(b) **Converges conditionally.** $1/\sqrt{n}$ decreases to 0, so the alternating series test gives convergence. But $\sum 1/\sqrt{n}$ is a p-series with $p = \frac{1}{2} \leq 1$, so it diverges, and the convergence is not absolute.

(c) **Diverges.** The terms are positive and decreasing. By the integral test, $\int_2^X \frac{dx}{x \ln x} = \ln\ln X - \ln\ln 2 \to \infty$. (Or by Cauchy condensation: $\sum 2^k \cdot \frac{1}{2^k k \ln 2} = \frac{1}{\ln 2}\sum \frac{1}{k}$ diverges.)

(d) **Converges absolutely.** $|\sin n/n^2| \leq 1/n^2$, and $\sum 1/n^2$ converges, so the comparison test applies.

Marking. 1 point per part, for the correct verdict *and* a valid reason.

B5

Definition. f is uniformly continuous on E if for every $\varepsilon > 0$ there is $\delta > 0$ such that for all $x, y \in E$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$. Here δ depends only on ε, not on x.

On $[0,1]$*:* $|x^2 - y^2| = |x + y|\,|x - y| \leq 2|x - y|$. Take $\delta = \varepsilon/2$.

Not on $\mathbb{R}$*:* take $\varepsilon = 1$. Given any $\delta > 0$, let $x = 1/\delta$ and $y = x + \delta/2$. Then $|x - y| < \delta$, but

$$|y^2 - x^2| = (y - x)(y + x) = \frac{\delta}{2}\left(\frac{2}{\delta} + \frac{\delta}{2}\right) = 1 + \frac{\delta^2}{4} > 1.$$

So no δ works for $\varepsilon = 1$.

Marking. 1 for the definition, 1 for $[0,1]$, 2 for $\mathbb{R}$ (1 if the idea "slopes become steep" is there without a correct quantifier argument).

B6

Mean Value Theorem. If f is continuous on $[a,b]$ and differentiable on (a,b), there is $c \in (a,b)$ with $f(b) - f(a) = f'(c)(b-a)$.

For $x \neq y$, apply the theorem to sin on the interval between them: $\sin x - \sin y = \cos(c)(x-y)$ for some c, so $|\sin x - \sin y| \leq |x-y|$. The case $x = y$ is trivial.

If $f' = 0$ on an interval I and $x < y$ are in I, then $f(y) - f(x) = f'(c)(y-x) = 0$. So f is constant on I.

Marking. 1 for the statement (hypotheses included), 1.5 for each application.

B7

(a) $f(x) = 0$ for $0 \leq x < 1$ and $f(1) = 1$.

(b) **No.** $\sup_{[0,1]} |f_n - f| \geq \sup_{0 \leq x < 1} x^n = 1$ for every n, so the sup does not tend to 0. (Alternatively: a uniform limit of continuous functions is continuous, and f is not.)

(c) **Yes.** $\sup_{[0,1/2]} |f_n - 0| = 2^{-n} \to 0$.

Marking. 1 for (a), 2 for (b), 1 for (c).

B8

(a) Let f be Riemann integrable on $[a,b]$.

FTC part 1: if $F(x) = \int_a^x f$, then F is continuous, and $F'(x) = f(x)$ at every point x where f is continuous.

FTC part 2: if G is differentiable on $[a,b]$ with $G' = f$, then $\int_a^b f = G(b) - G(a)$.

(b) Let $F(u) = \int_0^u e^{t^2}\,dt$. Since e^{t^2} is continuous, $F'(u) = e^{u^2}$. By the chain rule,

$$\frac{d}{dx} F(x^2) = 2x\, e^{x^4}.$$

(c) Every subinterval of a partition contains both rationals and irrationals, so on every subinterval $\sup D = 1$ and $\inf D = 0$. Every upper sum is therefore 1 and every lower sum is 0. The upper and lower integrals (1 and 0) differ, so D is not Riemann integrable.

Marking. 1.5 for (a), 1 for (b), 1.5 for (c).

Section C — Linear algebra (module 1.3)

C1

The characteristic polynomial is $(2-\lambda)^2 - 1 = (\lambda - 1)(\lambda - 3)$, so the eigenvalues are 3 and 1.

- $\lambda = 3$: eigenvector $(1,1)$, normalised to $\frac{1}{\sqrt{2}}(1,1)$.
- $\lambda = 1$: eigenvector $(1,-1)$, normalised to $\frac{1}{\sqrt{2}}(1,-1)$.

$$Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad A^n = QD^nQ^T = \frac{1}{2}\begin{pmatrix} 3^n+1 & 3^n-1 \\ 3^n-1 & 3^n+1 \end{pmatrix}.$$

Check with $n = 1$: this gives $\frac{1}{2}\begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = A$.

Marking. 1 for the eigenvalues, 1 for orthonormal eigenvectors, 1 for the factorisation, 1 for A^n.

C2

(a) If $T : V \to W$ is linear and V is finite-dimensional, then $\dim V = \dim \ker T + \dim \operatorname{im} T$.

(b) $\dim \operatorname{im} T = 3$, so $\dim \ker T = 5 - 3 = 2$.

(c) Let $n = \dim V$. T is injective $\iff$ $\ker T = \{0\}$ $\iff$ $\dim \ker T = 0$ $\iff$ $\dim \operatorname{im} T = n$ (by rank–nullity) $\iff$ $\operatorname{im} T = V$ (a subspace of full dimension is the whole space) $\iff$ T is surjective.

Marking. 1 for (a), 1 for (b), 2 for (c).

C3

Suppose not. Among all linearly dependent sets of eigenvectors with distinct eigenvalues, choose one, $v_1, \dots, v_k$ with eigenvalues $\lambda_1, \dots, \lambda_k$, with k as small as possible. Then $\sum c_i v_i = 0$ with every $c_i \neq 0$ (otherwise a smaller set would be dependent), and $k \geq 2$ because $v_1 \neq 0$. Apply $T - \lambda_k I$:

$$\sum_{i=1}^{k-1} c_i(\lambda_i - \lambda_k)v_i = 0.$$

Every coefficient $c_i(\lambda_i - \lambda_k)$ is non-zero, so $v_1, \dots, v_{k-1}$ is a smaller dependent set. This contradicts the minimality of k.

(An induction on k is equally good.)

Marking. 4: complete. 3: the key step (apply $T - \lambda_k I$) is present but the minimality or induction is not set up cleanly. 2: the case $k = 2$ only.

C4

$T(1) = 0$, $T(x) = 1$, $T(x^2) = 2x$. Writing the images as columns:

$$[T] = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

The matrix is upper triangular with zero diagonal, so the characteristic polynomial is λ^3 and the only eigenvalue is 0. The eigenvectors are the non-zero solutions of $p' = 0$, which are the non-zero constants. The eigenspace has dimension $1 < 3$, so T **is not diagonalisable**. Since $T^2 \neq 0$ ($T^2(x^2) = 2$) and $T^3 = 0$, the minimal polynomial is λ^3, and the Jordan form is a **single** 3×3 **Jordan block** with eigenvalue 0.

Marking. 1 for the matrix, 1 for the eigen-data, 1 for "not diagonalisable" with a reason, 1 for the Jordan form.

C5

$u_1 = \frac{1}{\sqrt{2}}(1,1,0)$.

$w_2 = v_2 - \langle v_2, u_1\rangle u_1 = (1,0,1) - \frac{1}{2}(1,1,0) = (\frac{1}{2}, -\frac{1}{2}, 1)$. Its norm is $\sqrt{3/2}$, so $u_2 = \frac{1}{\sqrt{6}}(1,-1,2)$.

The projection is $\langle e_3, u_1\rangle u_1 + \langle e_3, u_2\rangle u_2 = 0 + \frac{2}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}}(1,-1,2) = \left(\frac{1}{3}, -\frac{1}{3}, \frac{2}{3}\right)$.

Check: the residual $e_3 - \text{proj} = (-\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is orthogonal to both v_1 and v_2.

Marking. 2 for Gram–Schmidt, 2 for the projection.

C6

Let A be real symmetric, $Av = \lambda v$ with $v \in \mathbb{C}^n \setminus \{0\}$ and $\lambda \in \mathbb{C}$. Write $v^* = \bar{v}^T$. Then $v^*Av = \lambda\, v^*v = \lambda\|v\|^2$.

The scalar v^*Av is real: its complex conjugate is $v^TA\bar{v}$, which is a 1×1 matrix and so equals its transpose $\bar{v}^TA^Tv = v^*Av$, using $A^T = A$. Since $\|v\|^2 > 0$ is real, $\lambda = v^*Av/\|v\|^2$ is real.

Marking. 4: complete. 3: uses $A^* = A$ correctly but skips why v^*Av is real. 2: works with real v only (circular).

Section D — Abstract algebra (module 1.4)

D1

Lagrange. If H is a subgroup of a finite group G, then $|H|$ divides $|G|$.

Proof. The left cosets gH partition G: each g lies in gH, and if $gH \cap g'H \neq \emptyset$ then $gH = g'H$. Each coset has $|H|$ elements, because $h \mapsto gh$ is a bijection $H \to gH$. So $|G| = [G : H]\,|H|$.

Prime order. Let $|G| = p$, prime. Take $g \neq e$. Then $|\langle g\rangle|$ divides p and is greater than 1, so it equals p. Hence $G = \langle g\rangle$.

Marking. 1 for the statement, 2 for the proof (the partition and equal sizes), 1 for the corollary.

D2

$\mathbb{Z}/12$ is cyclic, so it has exactly one subgroup of each order d dividing 12, namely $\langle 12/d\rangle$:

Order	Subgroup
1	$\{0\}$
2	$\langle 6\rangle = \{0, 6\}$
3	$\langle 4\rangle = \{0, 4, 8\}$
4	$\langle 3\rangle = \{0, 3, 6, 9\}$
6	$\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$
12	the whole group

The generators are the residues coprime to 12: $1, 5, 7, 11$, which is $\varphi(12) = 4$ elements.

Marking. 3 for the subgroups (deduct 1 per missing one), 1 for the generators.

D3

(a) $\ker\varphi$ is a subgroup: $\varphi(e) = e$; if $\varphi(a) = \varphi(b) = e$ then $\varphi(ab^{-1}) = e$. It is normal: for $k \in \ker\varphi$ and $g \in G$, $\varphi(gkg^{-1}) = \varphi(g)\,e\,\varphi(g)^{-1} = e$.

(b) **First Isomorphism Theorem.** $\ker\varphi \trianglelefteq G$, and $G/\ker\varphi \cong \operatorname{im}\varphi$ via $g\ker\varphi \mapsto \varphi(g)$.

(c) $\varphi : \mathbb{R}^\times \to \mathbb{R}_{>0}$, $\varphi(x) = |x|$, is a homomorphism ($|xy| = |x||y|$). It is surjective ($\varphi(r) = r$ for $r > 0$), and its kernel is $\{\pm 1\}$. Apply (b).

Marking. 1.5 for (a), 1 for (b), 1.5 for (c).

D4

(a) **Yes.** A cubic over a field is reducible only if it has a linear factor, that is, a root. By the rational root test the only possible rational roots of $x^3 - 2$ are $\pm 1, \pm 2$, and none is a root ($1 - 2$, $-1 - 2$, $8 - 2$, $-8 - 2$ are all non-zero).

(b) $\mathbb{R}[x]$ is a principal ideal domain, and there a non-zero ideal (p) is maximal if and only if p is irreducible. $x^2 + 1$ has degree 2 and no real root, so it is irreducible over $\mathbb{R}$. Hence $(x^2 + 1)$ is maximal and the quotient is a field.

Evaluation $\mathrm{ev}_i : \mathbb{R}[x] \to \mathbb{C}$, $p \mapsto p(i)$, is a ring homomorphism. It is surjective, since $a + bx \mapsto a + bi$. Its kernel contains $x^2 + 1$, and since the kernel is a proper ideal containing the maximal ideal $(x^2 + 1)$, it equals $(x^2 + 1)$. So $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.

Marking. 1 for (a), 3 for (b): 1 for maximality, 2 for the identification with $\mathbb{C}$.

D5

(a) $7 \cdot 3 = 21 \equiv 1 \pmod{20}$, so $7^{-1} \equiv 3$, and $x \equiv 3 \cdot 3 = 9 \pmod{20}$. Check: $63 = 60 + 3$.

(b) Powers of 3 modulo 7: $3, 2, 6, 4, 5, 1$. The order is $6 = |(\mathbb{Z}/7)^\times|$, so 3 generates the group, which is therefore cyclic.

Marking. 2 per part.

Section E — Multivariable calculus (module 1.5)

E1

$f(x, 0) = 0$ for all x, so $f_x(0, 0) = 0$. Likewise $f_y(0, 0) = 0$. Both partial derivatives exist.

Along the line $y = x$, $f(t, t) = \frac{1}{2}$ for $t \neq 0$, which does not tend to $f(0, 0) = 0$. So f **is not continuous** at the origin, and therefore **not differentiable** there (differentiability implies continuity).

Lesson: the existence of partial derivatives does not imply differentiability, or even continuity.

Marking. 1 for the partials, 2 for discontinuity, 1 for non-differentiability with its reason.

E2

$f_x = 3x^2 - 3$ and $f_y = 2y$, so the critical points are $(1, 0)$ and $(-1, 0)$. The Hessian is $\begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}$.

- At $(1, 0)$: $\mathrm{diag}(6, 2)$ is positive definite, so this is a **local minimum**, with $f = -2$.
- At $(-1, 0)$: $\mathrm{diag}(-6, 2)$ is indefinite, so this is a **saddle point**.

Marking. 2 for the critical points, 1 per classification.

E3

With $g(x,y) = x^2 + 4y^2$, the condition $\nabla f = \lambda \nabla g$ gives $y = 2\lambda x$ and $x = 8\lambda y$. Hence $y = 16\lambda^2 y$.

If $y = 0$ then $x = 0$, which is not on the ellipse. So $\lambda^2 = 1/16$, and $x = 8\lambda y = \pm 2y$. The constraint gives $4y^2 + 4y^2 = 8$, so $y = \pm 1$.

The candidate points are $(2,1)$ and $(-2,-1)$, where $f = 2$, and $(2,-1)$ and $(-2,1)$, where $f = -2$. The ellipse is compact and f is continuous, so the extremes are attained, and they must be among these candidates. **The maximum is 2 and the minimum is −2.**

Marking. 1 for the equations, 2 for solving them, 1 for the conclusion (compactness remark optional).

E4

In polar coordinates, with $dA = r\,dr\,d\theta$:

$$\int_0^{2\pi} \int_0^R e^{-r^2} r\,dr\,d\theta = 2\pi \cdot \frac{1 - e^{-R^2}}{2} = \pi\left(1 - e^{-R^2}\right).$$

Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$. It converges, by comparison with $e^{-|x|}$ for $|x| \geq 1$. By Fubini, $I^2 = \lim_{R\to\infty} \iint_{[-R,R]^2} e^{-(x^2+y^2)}\,dA$. The square $[-R,R]^2$ lies between the disks of radius R and $R\sqrt{2}$, and the integrand is positive, so

$$\pi\left(1 - e^{-R^2}\right) \leq \iint_{[-R,R]^2} e^{-(x^2+y^2)}\,dA \leq \pi\left(1 - e^{-2R^2}\right).$$

Let $R \to \infty$: $I^2 = \pi$. Since $I > 0$, $I = \sqrt{\pi}$.

Marking. 2 for the disk integral (including the Jacobian r), 2 for the deduction. Give 1 for the deduction if the squeeze between disk and square is omitted.

E5

Directly: with $x = a\cos t$, $y = b\sin t$, $0 \leq t \leq 2\pi$,

$$-y\,dx + x\,dy = \left(ab\sin^2 t + ab\cos^2 t\right) dt = ab\,dt, \quad \text{so the integral is } 2\pi ab.$$

Green's theorem: with $P = -y$ and $Q = x$, $\oint_C P\,dx + Q\,dy = \iint_{\text{inside}} (Q_x - P_y)\,dA = 2 \cdot \text{Area} = 2\pi ab$.

Marking. 2 per method.

Section F — Complex numbers and topology basics

F1

$-16 = 16e^{i\pi}$. So $z = 2e^{i(\pi/4 + k\pi/2)}$ for $k = 0,1,2,3$. These are the angles $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$. In Cartesian form: $\pm\sqrt{2} \pm i\sqrt{2}$ (all four sign combinations).

Marking. 2 for the polar form, 2 for the Cartesian form. Deduct 1 for each missing root.

F2

(a) $\bar{z} = x - iy$, so $u = x$ and $v = -y$. The Cauchy–Riemann equation $u_x = v_y$ would need $1 = -1$, which never holds. So $\bar{z}$ **is complex differentiable nowhere**.

(b) With $z = e^{it}$, $dz = ie^{it}dt$, the integral is $\int_0^{2\pi} \frac{ie^{it}}{e^{it}}\, dt = 2\pi i$.

Marking. 2 per part.

F3

(a) $\bigcap_{n\geq 1}(-\frac{1}{n}, \frac{1}{n}) = \{0\}$, which is not open: no interval around 0 is contained in it.

(b) **Heine–Borel.** A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

(c) The open sets $U_n = (\frac{1}{n}, 2)$ for $n \in \mathbb{N}$ cover $(0, 1]$, because every $x > 0$ has $x > 1/n$ for some n. Any finite subfamily has a largest index N, and its union is $(\frac{1}{N}, 2)$, which misses $\frac{1}{2N} \in (0, 1]$. So there is no finite subcover.

Marking. 1 for (a), 1 for (b), 2 for (c).

Score sheet

Copy this into PROGRESS.md.

Section	Score	Max	%	Route (see README)
A (1.1)		24		
B (1.2)		32		
C (1.3)		24		
D (1.4)		20		
E (1.5)		20		
F (2.1 and 2.2 preview)		12		informational only
Total		**132**		